

Task-2: Network Scanning & Vulnerability Assessment

Nmap Scan Report

1. Objective

The objective of this task was to perform network reconnaissance, port scanning, service detection, operating system identification, and vulnerability assessment on the Metasploitable target machine using Nmap. The testing was conducted inside a controlled virtual lab environment.

2. Target Information

- Target Machine: Metasploitable 2
 - Target IP Address: <target-ip>
 - Attacker Machine: Kali Linux
 - Network Type: Host-Only Adapter (Isolated Lab Network)
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3. Host Discovery

To verify that the target machine was active and reachable, the following command was used:

```
nmap -sn <target-ip>
```

The result confirmed that the host was up and responding to network requests.

4. Basic Port Scanning

A standard TCP port scan was performed using:

```
nmap <target-ip>
```

The scan identified multiple open ports, including:

- 21/tcp – FTP
- 22/tcp – SSH
- 23/tcp – Telnet
- 25/tcp – SMTP
- 80/tcp – HTTP
- 3306/tcp – MySQL

These open ports indicate active services running on the target system.

5. SYN (Stealth) Scan

A SYN scan was performed to identify open ports using a half-open TCP handshake:

```
nmap -sS <target-ip>
```

This scan confirmed the previously detected open ports while minimizing full TCP connections.

6. Service Version Detection

To identify the versions of services running on the open ports, the following command was executed:

```
nmap -sV <target-ip>
```

The scan revealed service details such as:

- vsFTPD 2.3.4
- OpenSSH
- Apache HTTP Server
- MySQL Database Service

Identifying service versions is important for detecting known vulnerabilities associated with outdated software.

7. Operating System Detection

To determine the operating system of the target machine, the following command was used:

```
nmap -O <target-ip>
```

The results indicated that the target system is running a Linux-based operating system, consistent with the Metasploitable configuration.

8. Vulnerability Scanning Using Nmap NSE Scripts

A vulnerability assessment was performed using Nmap's built-in vulnerability scripts:

```
nmap -sS -sV -O --script vuln <target-ip>
```

This scan analyzed open services against known Common Vulnerabilities and Exposures (CVEs).

The results identified multiple security weaknesses due to outdated software versions and insecure service configurations. Several vulnerabilities were associated with FTP, HTTP, and database services running on the target machine.

9. Observations

- The target machine exposes multiple open ports.
- Several services are outdated and potentially vulnerable.

- The system presents a large attack surface due to unnecessary services running.
 - Vulnerability scripts confirmed the presence of known security issues.
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10. Conclusion

The Nmap scanning process successfully identified open ports, running services, operating system details, and potential vulnerabilities on the target machine.

The results demonstrate how improper service configuration and outdated software can increase security risks. This task provided practical experience in network reconnaissance, scanning techniques, and vulnerability assessment within a controlled lab environment.